

Balochistan University of Engineering and Technology, Khuzdar



CONTINUOUS QUALITY IMPROVEMENT FRAMEWORK

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Continuous Quality Improvement (CQI)

Continuous Quality Improvement (CQI) in a university's Outcome-Based Education (OBE) system is a data-driven, cyclical process used to monitor, evaluate, and enhance student learning and program performance over time.

Core Levels of CQI in OBE

CQI operates across three interconnected tiers to ensure comprehensive academic enhancement:

1. Course Level (CLO Cycle):

- Focuses on Course Learning Outcomes.
- Conducted at the end of every semester by each course teacher.
- Involves analyzing assignment, quiz, and exam scores mapped to specific learning outcomes.
- Adjusts teaching methods, lab exercises, or assignment weights if student attainment falls below target benchmarks.

2. Program Level (PLO Cycle):

- Tracks Program Learning Outcomes across multiple courses to evaluate graduating competencies.
- Evaluates whether collective course achievements fulfill broader graduation requirements.
- Refines curriculum sequencing, prerequisite structures, or faculty development needs.

3. Program Objective Level (PEO Cycle):

- Measures Program Educational Objectives 3 to 5 years after graduation.
 - Gathers indirect feedback via alumni surveys, employer evaluations, and industry advisory panels.
 - Guides major, long-term curriculum overhauls to match shifting workforce demands
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Outline

CLO CQI process/cycle

Key Performance Indicators (KPI) of CLOs

CLO Corrective Actions

PLO CQI process/cycle

Key Performance Indicators (KPI) of PLOs

PLO Corrective Actions

PEO CQI process/cycle

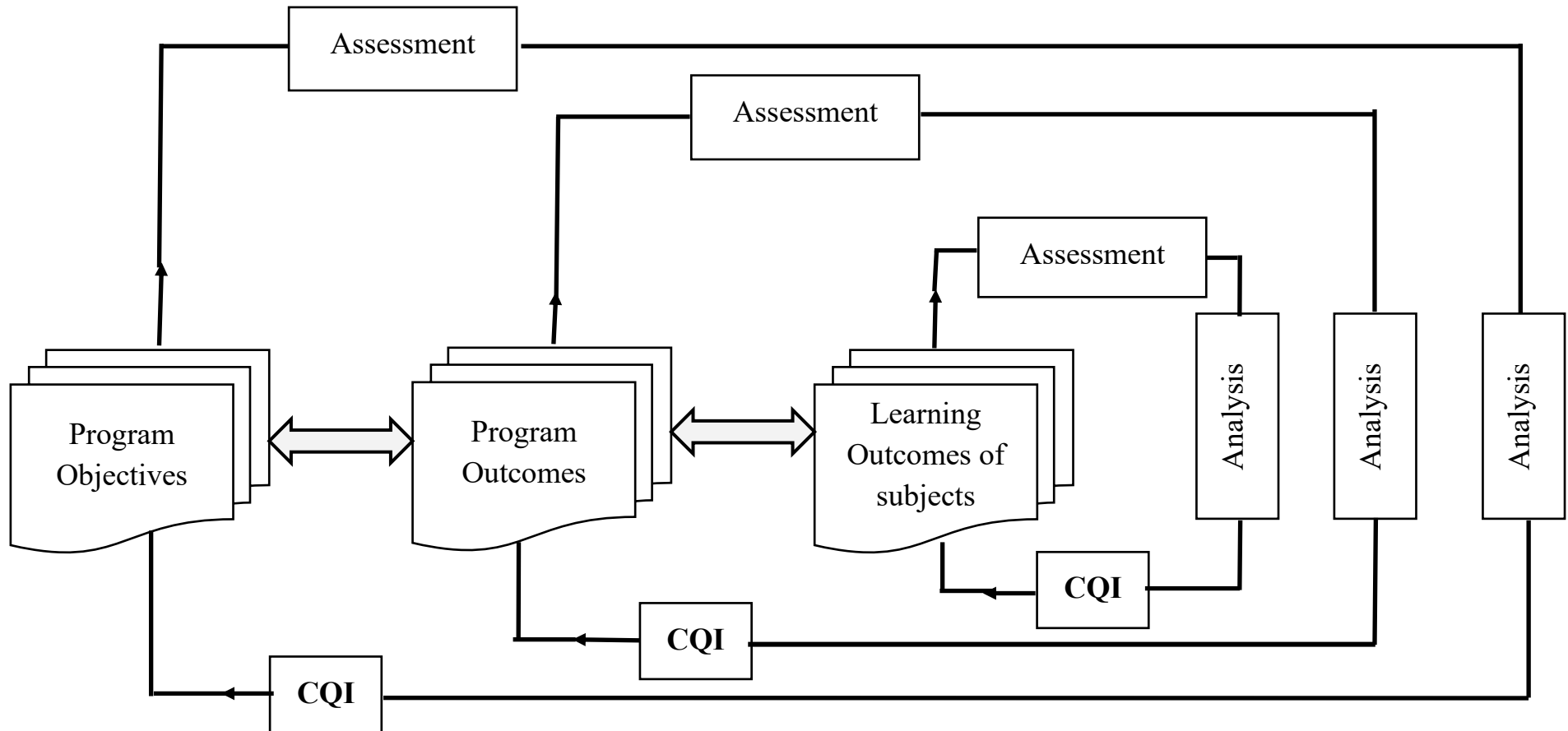
Key Performance Indicators (KPI) of PEOs

PEO Corrective Actions

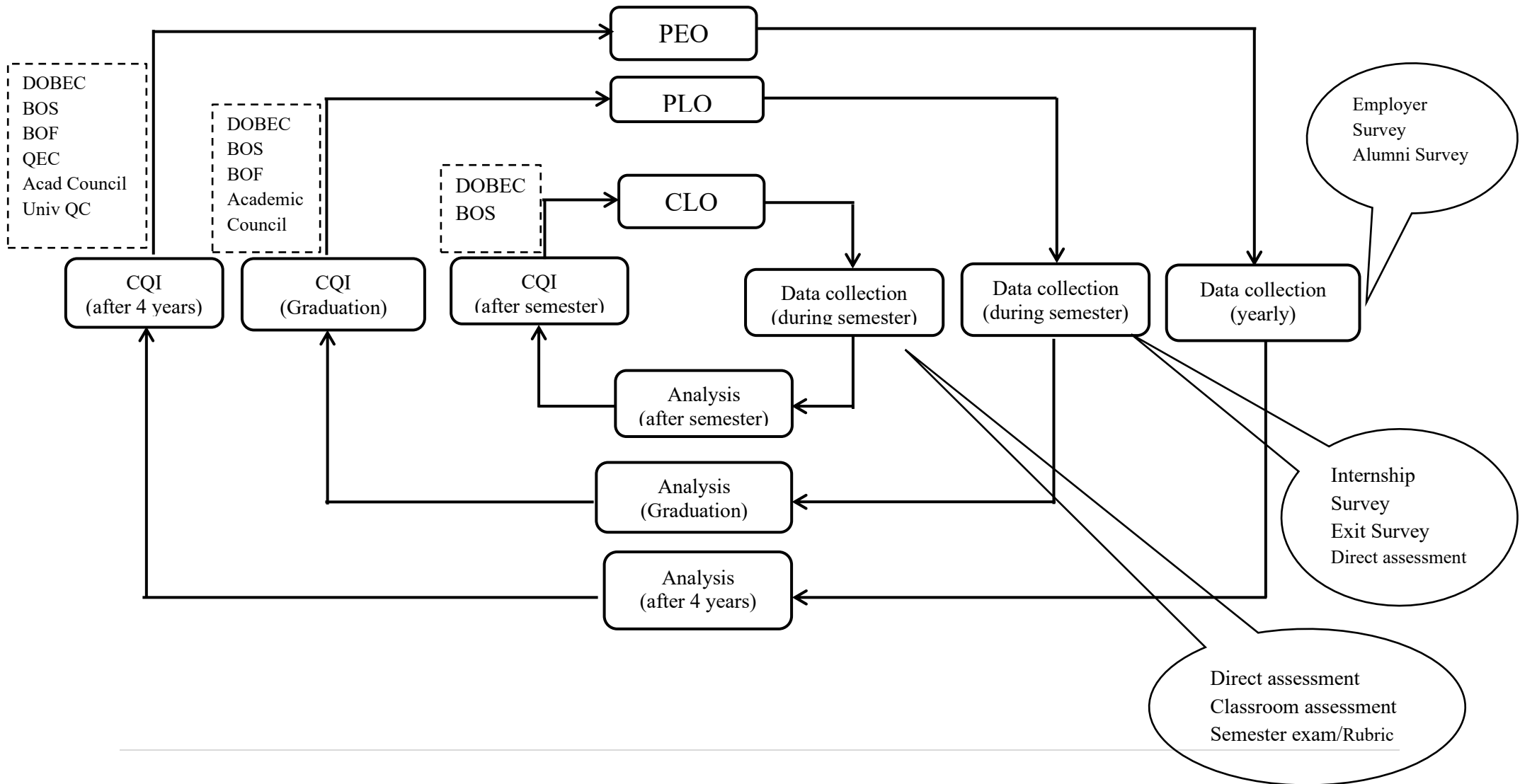
Survey Forms/Assessment tools in OBE System

OBE Organ	Type of Survey	Data Collection	Analysis	CQI	Sources	To be done by
PEOs	1. Employer's Survey 2. Alumni Survey	Yearly	4 Yearly	4 yearly	Industry Alumni	QEC QEC
PLOs	1. Exit Survey 2. Internship Feedback	Semester	Graduation	Graduation	Student Industry	QEC ORIC/Dept
CLOs	1. Affective Rubrics 2. Quiz/CEP/OEL/Exams	Semester	Semester	Semester	Student	Faculty Faculty
Course Evaluation	1. Course Evaluation Survey	Semester	Semester	Semester	Student/ Faculty	QEC
Faculty Evaluation	1. Faculty Evaluation Survey	Semester	Semester	Semester	Student	QEC

CQI Cycle



CQI Process

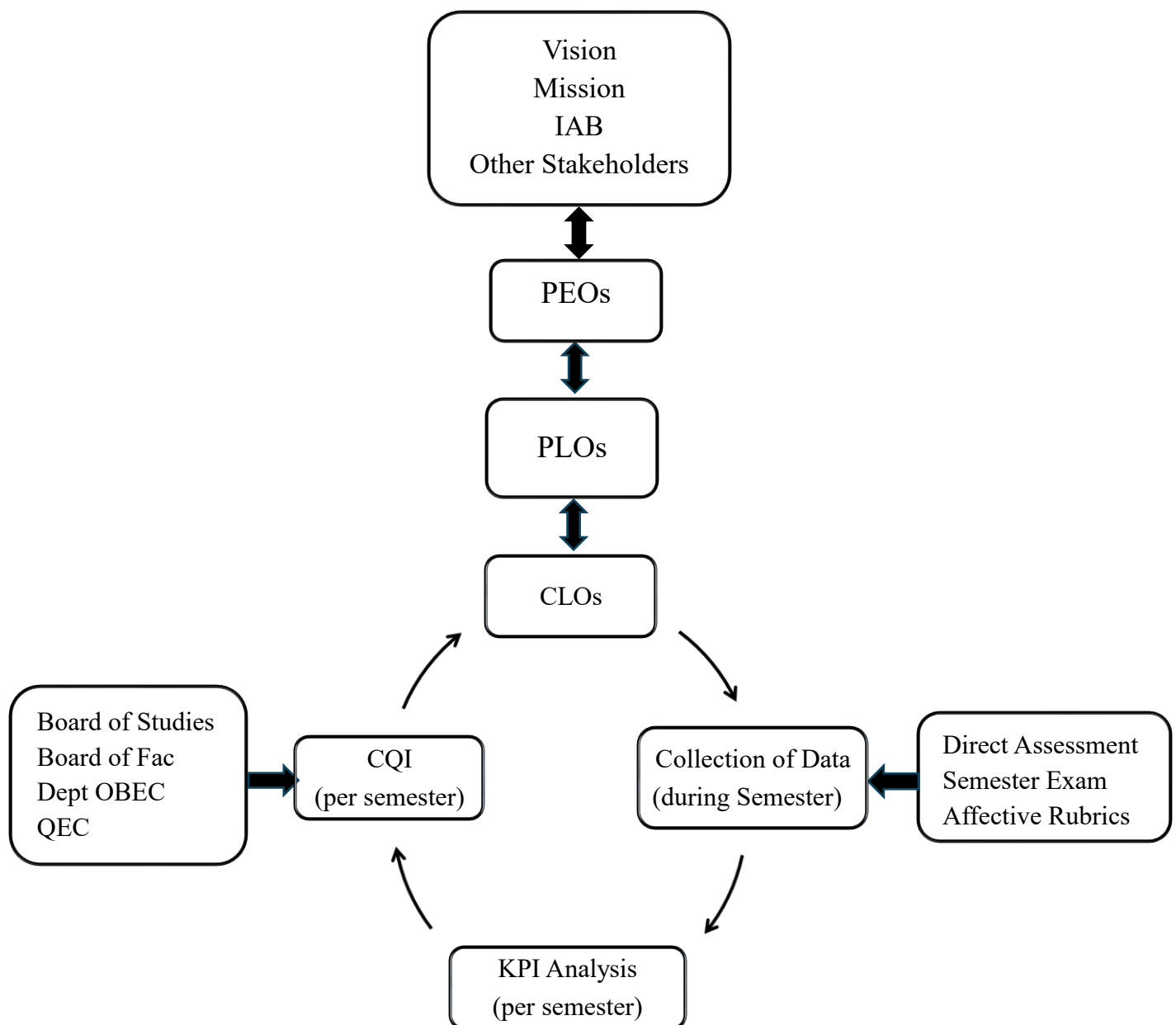


Assessment Tools

Assessment Tool		Objectives of assessment	Assessment frequency
Direct	Course assessment tools	To determine the students' achievement in the courses taken. Assessment is carried out through quizzes, tests, assignments, CEP project/laboratory presentations, OEL/laboratory reports and final examinations.	Every semester
Indirect	End of Course Survey	To acquire students' opinions of the achievement of the COs for every course	Every semester
	Exit Survey	To determine the students' views on the program and their assessment on the overall achievement of the PLOs.	Every year
	Industrial Training Employer Survey	To determine the students' performance during their Industrial training attachment and to assess the PEO	Every year
	Alumni Survey	To seek for the alumni's opinion on the suitability and relevance of the PEOs to their current employment	Every 3-4 years
	Industrial Advisory Panel feedback	To determine the strength and weaknesses of the program from an industry point of view. To identify the needs of Industry	Every year

1. Course Learning Outcome (CLO) CQI Process

Course Learning Outcomes are specific and measurable statements that define the knowledge, skills, and attitudes learners will demonstrate by the completion of a course. Learning Outcomes are written with a verb phrase and declare a demonstrable action within a given time frame, such as by the end of the semester. Ideally, they should be observable, measurable, and achievable within a specified time period. More specially they should be SMART.



1.1 Key Performance Indicator (KPIs) of CLOs

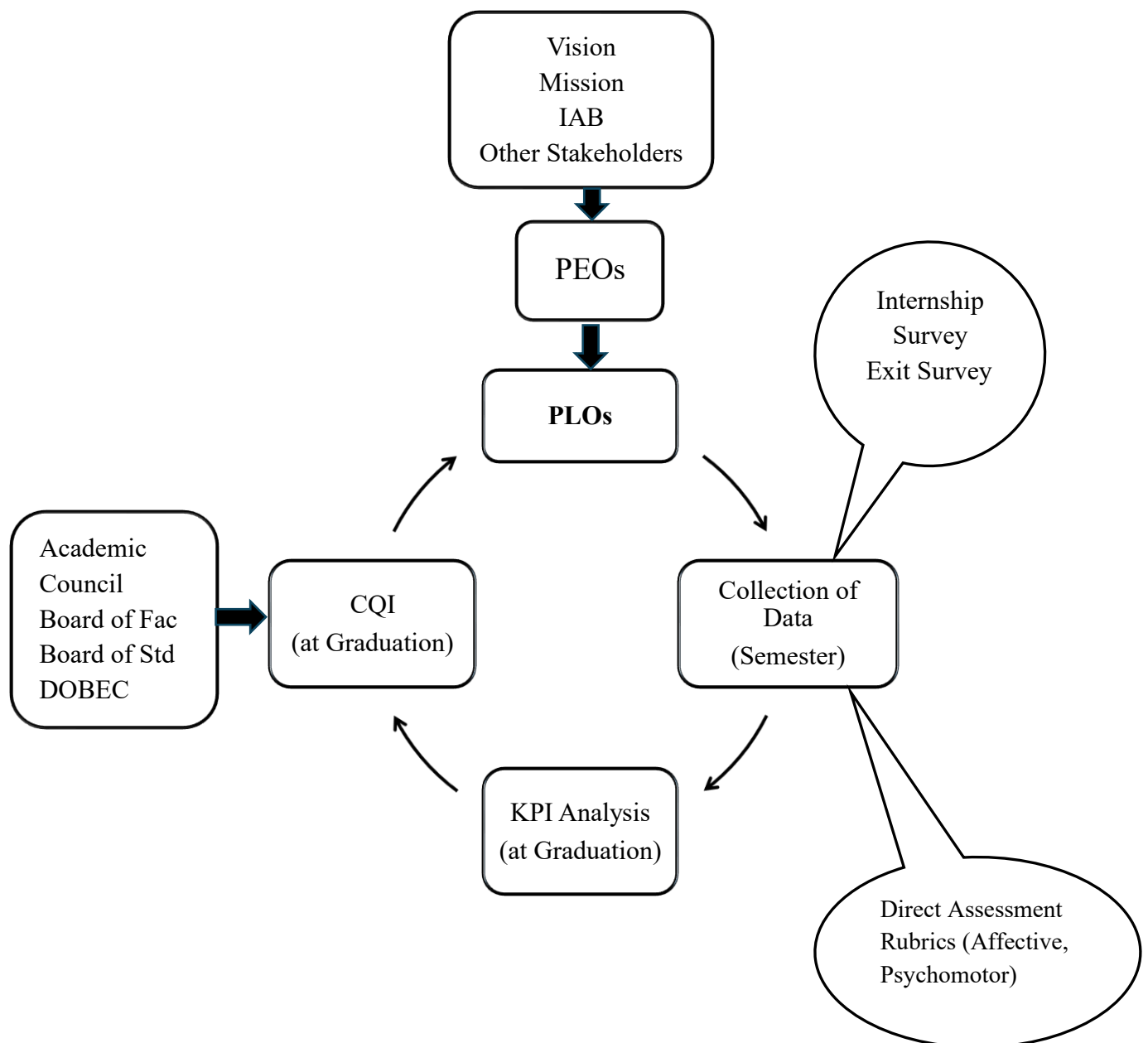
Measurement Tool	KPIs	Frequency	Objective of assessment
Direct assessment Methods (Assignments, quiz, midterm, final term, lab exam)	i. Individual – A student attains a CLO by getting 50% of the marks assigned to that CLO ii. Cohort – At least 50% of the students pass the CLO	At the end of the semester	To measure the effectiveness of knowledge acquired in a course. To know that effective knowledge has been imparted to the students.

1.2 Corrective Actions of CLOs

Individual KPIs Achieved	Cohort KPIs Achieved
• Set New Target	• Set New Target
Individual KPIs Not Achieved	Cohort KPIs Not Achieved
1. Opportunity to reappear in supplementary exam, winter camp, and he/she can register for course improvement. 2. Arrangement of special tutor for weak students	1. Course reassessment 2. Review of CLOs assessment methods 3. Faculty advice by Chairman 4. Faculty development/training 5. Review/revision of course content 6. Revision of CLOs 7. Review of taxonomy level 8. Review of assessment methodology 9. Review of other course-related facilities 10. Review of laboratory & equipment facilities

2. Program Learning Outcome (PLO) CQI Process

Program Learning Outcomes (PLOs) represent the attributes an engineering graduate is expected to achieve by the time of graduation. They encompass the knowledge, skills, and attitudes developed throughout the degree program. To make it simplify, PLOs are clear statements that describe the knowledge, skills, values, and competencies students must demonstrate by the time they graduate from an academic program.



2.1 Key Performance Indicator of PLOs

Measurement Tool	KPIs	Frequency	Objective of assessment
Mapped courses CLO assessment (Direct)	1. Individual – A student attains a PLO by passing 50% of the mapped courses 2. Cohort (program) – At least 50% of the students have attained the PLO in each mapped course	Upon graduation	To determine the students' actual achievement in the courses taken. Assessment is carried out through quizzes, tests, assignments, project/laboratory presentations, and final examinations.
Internship survey form	Cohort – Minimum 50% of responses carry at least half the consolidated score for a PLO	After internship	To determine the students' performance during their Industrial Training attachment and to assess the PLOs.
Exit Survey form	Cohort- Minimum 50% of responses carry at least half the consolidated score for a PLO	Upon graduation	To determine the student views on the program and their assessment on the overall achievement of the PLOs.

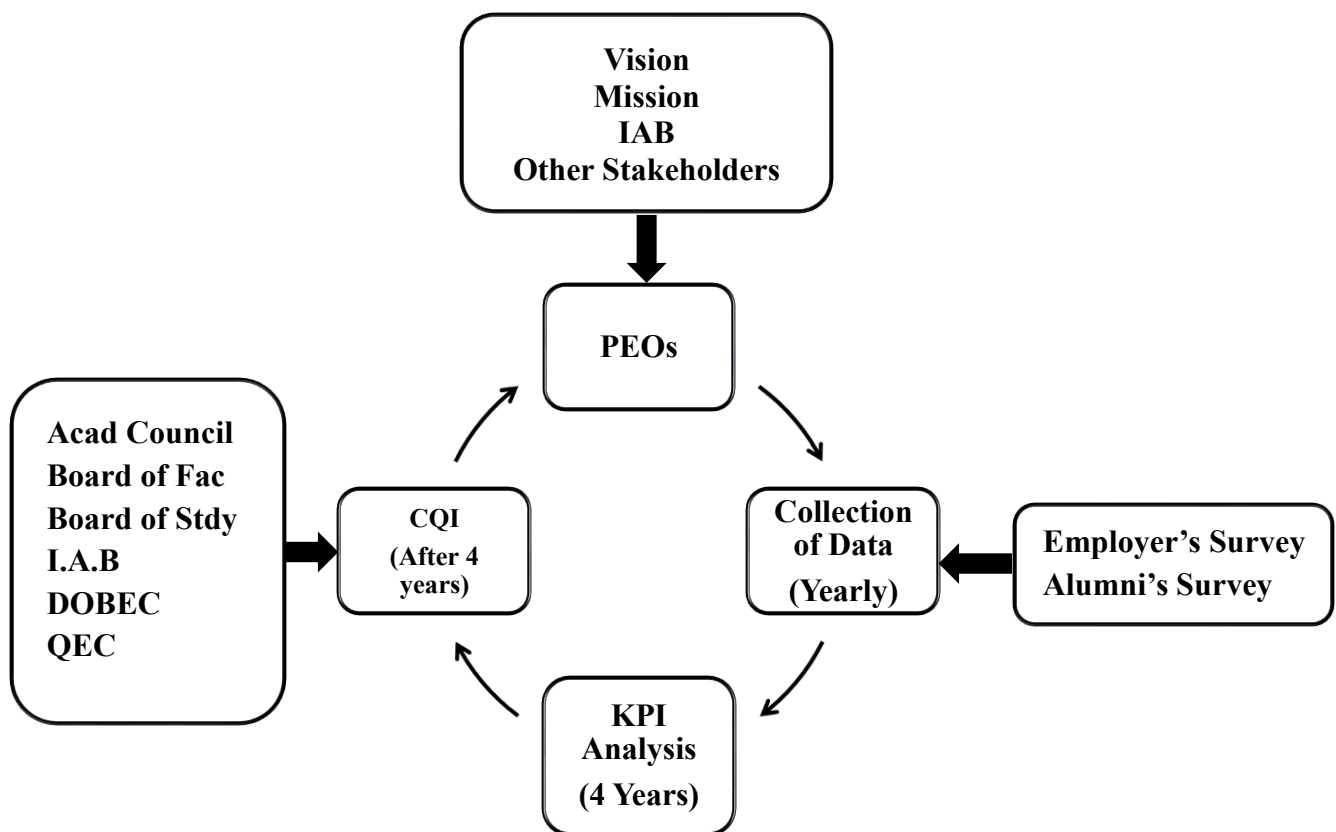
2.2 Corrective Actions of PLOs

Individual KPIs Achieved
3. No action required
Individual KPIs Not Achieved
To make students aware of their progress, three critical stages are defined below with appropriate actions. 1. Counseling Stage - At the end of each semester, the academic branch will publish individual Program Learning Outcomes (PLOs) attainment along with semester results. - If a student fails to attain the assessed PLO, counseling will be conducted by the appointed academic advisor of the class. - During counseling sessions, advice on the prospects of covering that PLO in subjects that are yet to be offered will be provided to the student. - A Counseling Performa will be completed during the counseling session to maintain a record of the discussion. - The completed Counseling Performa will be placed in the dossier of the concerned student for future reference and tracking of progress. 2. Warning Stage - If a student fails to achieve the defined level of Program Learning Outcome (PLO) attainment as per Key Performance Indicator (KPI) after it has been offered more

than 50% of the total times: Written warning letters will be issued by the exam branch/HoD through the respective academic advisor. ▪ The purpose of these warning letters is to encourage the student to focus on and recover the failed PLOs in upcoming courses. ▪ The issued warning letter is maintained in the dossier file of the concerned student for documentation and tracking purposes.
Cohort KPIs Achieved
4. Set new targets
Cohort (Program) KPIs Not Achieved
• Review of teaching and learning process • Review of PLOs assessment methods • Review of CLO-PLO mapping • Review of curriculum design • Review of relevant KPIs • Review of implementation of CARs

3. Program Education Outcome (PEO) CQI

Program Educational Objectives (PEOs) are broad statements that define the career and professional achievements graduates are expected to attain a few years after completing a degree program. PEOs must be consistent with the vision and mission of the university and the faculty and are intended to satisfy the needs of the stakeholders.



3.1 KPI of PEOs

Measurement Mechanism	PEOs Measured	KPI	Attributes Covered
Employer's feedback form	PEO 1, 2, 3	Minimum 50% of responses carry at least half the consolidated score for a PEO	1. PLO 1 2. PLO 2 3. PLO 3 4. PLO 4 5. PLO 5 6. PLO 6 7. PLO 7 8. PLO 8 9. PLO 9 10. PLO 10 11. PLO 11 12. PLO 12
Alumni's survey	PEO 1, 2, 3	Minimum 50% of responses carry at least half the consolidated score for a PEO	

3.2 PEO Corrective Actions

KPIs Achieved	KPIs Not Achieved
Set new targets	1. Review of teaching and learning process 2. Review of employer and alumni feedback survey forms 3. Review of PEOs 4. Review of curriculum strategies 5. Review of relevant KPIs 6. Review of implementation of CARs (corrective action reports)